Sachin J

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.98
maharishi international residential school	Class XII (CBSE)	2023	91.00%
queen mira international school	Class X (CBSE)	2021	89.00%

INTERNSHIP EXPERIENCE

Quantum Engineering Intern | Accelequant | Bangalore

- Engineered custom MIP and non-convex QP/QCQP presolve modules in Python and SCIP, incorporating research-backed bound tightening and constraint aggregation to drastically reduce model size across large-scale production benchmarks.
- Developed hybrid quantum-classical callback pipelines in Gurobi and SCIP that extract active search-tree subproblems, reformulate them into QUBO/Graph Partitioning structures, and inject primal feasible solutions to accelerate optimization convergence.
- Built end-to-end benchmarking infrastructure and testcase generators (QLS) to evaluate open-source solvers (PaPiLo, HiGHS, Minotaur) against commercial standards on production models scaling up to 2.95M variables.

SCHOLASTIC ACHIEVEMENTS

Codeforces Expert (Max Rating: 1673): Solved **1,500+** algorithmic problems across Codeforces and LeetCode.

Google Code to Learn (2018): Qualified the nationwide App Inventor assessment.

SKILLS

Languages - C, C++, Python, Java, HTML, CSS, JavaScript, Node.js, MongoDB.

Libraries & Frameworks - NumPy, C++ STL, Jupyter Notebook, React, Express.

Developer Tools - Git, GitHub, VS Code, LaTeX.

RELEVANT COURSES

Computer Science:

- Introduction to Programming, Data Structures and Algorithms, Algorithms, Operating Systems I, Software Development Fundamentals, Discrete Mathematics, Introduction to Computing Core

Mathematics:

- Calculus I, Calculus II, Introduction to Statistics, Introduction to Probability, Linear Algebra

Applied Mathematics:

- Probability and Random Variables, Transform Techniques, Differential Equations, Numerical Methods, Complex Variables

PROJECTS

Go Clean Architecture Backend | Personal Project | GitHub

- Architected a high-throughput **REST API** in **Go** using **Clean Architecture** to decouple business logic from **MongoDB** and enable mock-driven testing via **Testify** and **Mockery**.

- Implemented stateless authentication using **JWT Access and Refresh Tokens**, building custom **Gin middleware** for token verification, payload decoding, and route sandboxing.
- Engineered secure credential storage via adaptive **bcrypt password hashing**, **Viper** environment pipelines, and structured error handling to isolate database internals from JSON payloads.
- Configured **multi-stage Docker builds** and **Docker Compose** to orchestrate local dependencies, enabling zero-downtime **MongoDB Atlas** cloud connectivity and reproducible execution.

Arbitrary Precision Arithmetic Library

Course: *Software Development Fundamentals* | **Instructor**: Prof. Ashish Mishra

- Developed a **Java-based library** for arbitrary-precision arithmetic, supporting operations on large integers and floating-point numbers.
- Designed modular, object-oriented classes (**AInteger**, **AFloat**) with array-based algorithms for **addition**, **subtraction**, **multiplication**, and **division**.
- Implemented a **Python CLI wrapper** and integrated an **Ant build system** for cross-platform automation and usability.
- Applied concepts from **numerical methods** and **precision control**, and demonstrated effective integration of **Java and Python**.

EXTRACURRICULAR

Basketball

- Represented the state in the **State-Level Basketball Tournament**, held in **Tamil Nadu (2019)**.

Cycling

- Awarded **silver medal** in the **World Cyclist Day Competition**, held at **IIT Hyderabad**.

Taekwondo

- Won **two gold**, **three silver**, and **three bronze** medal in **State-Level Taekwondo Championships (2019, 2020)**

POSITION OF RESPONSIBILITY

Core Member, EBSB Club, IIT Hyderabad

- Organized cross-cultural events and interactive workshops for **[100+]** students to promote inter-state cultural exchange.